M^2-CL: Multi-Scale and Multi-Layer Contrastive Learning for Domain Generalization

Reproduction Report

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Abstract

Domain generalization (DG) aims to train models on multiple source domains such that they generalize to unseen target domains without any target data during training. This paper proposes **M^2-CL**, a framework that improves DG by simultaneously exploiting multi-scale and multi-layer feature representations extracted from a standard ResNet backbone. Extraction blocks attached to 13 intermediate convolutional layers apply parallel concentration pipelines with spatial dropout and multi-scale max pooling. A novel multi-layer contrastive objective enforces domain-invariant alignment of same-class embeddings at every layer. M^2-CL achieves state-of-the-art results on PACS, VLCS, Office-Home, and NICO benchmarks.

1. Introduction

Deep convolutional neural networks are known to exploit domain-specific cues during training, leading to poor generalization when the test distribution differs from the training distribution. Domain generalization addresses this challenge by learning representations that are invariant to domain shifts without ever observing the target domain.

Existing approaches tackle DG via data augmentation, meta-learning, domain alignment, or contrastive objectives. Most methods, however, rely exclusively on the final feature representation of the backbone — discarding the rich multi-level information encoded in intermediate layers.

M^2-CL addresses this gap by attaching lightweight extraction blocks to 13 intermediate convolutional layers of a ResNet-18 backbone. Each block applies parallel concentration pipelines that pool features at multiple spatial scales, generating a compact normalized embedding $u^{(l)}$ per layer. A multi-layer contrastive loss then enforces that embeddings of same-class samples are aligned at every layer, regularizing the backbone toward domain-invariant representations.

2. Related Work

2.1 Domain Generalization

Standard approaches include empirical risk minimization (ERM), domain-invariant feature learning via MMD or CORAL alignment, data augmentation (Mixup, RSC), and meta-learning strategies (MAML-based). DomainBed provides a unified evaluation framework for fair comparison across these methods.

2.2 Contrastive Learning

Self-supervised contrastive learning (SimCLR, MoCo) learns representations by attracting augmented views of the same image and repelling views from different images. Supervised contrastive learning extends this by using class labels to define positive pairs. M^2-CL adapts supervised contrastive learning to the domain generalization setting by applying it across M intermediate layers simultaneously.

3. Proposed Method: M²-CL

3.1 Extraction Block

An extraction block is attached to each of the 13 intermediate convolutional layers of ResNet-18. Given a feature map **F** ∈ ■$^{C\times H\times W}$, the block applies:

(i) A 1×1 convolution that reduces channels by reduction ratio *r* (default 4), followed by BatchNorm and ReLU.
(ii) Spatial Dropout (Dropout2d) that drops entire feature maps with probability $p = 0.5$, forcing the model to build distributed representations.
(iii) Parallel concentration pipelines with AdaptiveMaxPool2d at multiple output sizes. Early layers use sizes {8×8, 4×4, 2×2}; later layers use {7×7, 3×3}.
(iv) Each pooled map is flattened and passed through a linear MLP to produce a fixed-size embedding vector.
(v) All pipeline outputs are concatenated and projected to a final embedding, then L2-normalized to produce $\mathbf{u}^{(l)}$ ∈ ■^{128}.

3.2 Multi-Layer Contrastive Loss

For each layer *l*, the probability measure for class *c* is defined as:

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p^(l)(c) = Σ_{i,j: y_i=y_j=c} exp( u_i^(l)■ u_j^(l) / τ )
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Σ_{k,m: k≠m} exp( u_k^(l)■ u_m^(l) / τ )
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L^(l) = -Σ_c log p^(l)(c)

L = L_CE + α · Σ_{l=1}^{M} L^(l)
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where τ is the temperature (default 1.0), α controls the relative importance of the contrastive term (default 0.01), and $M = 13$ is the number of extraction blocks.

4. Experimental Setup

4.1 Datasets

Dataset	Domains	Images	Classes	Protocol
PACS	4	9,991	7	Leave-one-domain-out
VLCS	4	10,729	5	Leave-one-domain-out
Office-Home	4	15,588	65	Leave-one-domain-out
NICO	**Variable**	**25,000**	**19**	**Hold-out N contexts**

Table 1. Benchmark datasets used for evaluation.

4.2 Training Details

All experiments use a ResNet-18 backbone pre-trained on ImageNet. Training uses SGD with momentum 0.9, initial learning rate 0.001, cosine annealing schedule, batch size 128, and 30 epochs. Data augmentation follows DomainBed: random resized crop to 224×224, random horizontal flip, color jitter, and occasional grayscale. Results are averaged over 3 random seeds. Hardware: NVIDIA RTX A5000.

5. Results

5.1 PACS

Method	Art	Cartoon	Photo	Sketch	Avg
ERM	77.37	75.51	96.11	69.27	84.51
RSC	75.21	74.68	95.84	75.03	80.11
CORAL	76.23	77.01	93.58	70.35	84.29
Mixup	78.11	73.67	94.29	74.75	84.13
SagNet	77.55	76.41	95.16	77.89	83.58
SelfReg	77.81	76.35	95.42	74.32	80.97
SAGM	79.21	77.35	94.61	79.58	82.68
M²-CL	**81.66**	**78.42**	**97.00**	**77.07**	**83.54**

Table 2. PACS results (ResNet-18, top-1 accuracy %). M²-CL row highlighted.

5.2 VLCS

Method	PASCAL	LabelMe	Caltech	SUN09	Avg
ERM	72.11	62.14	95.47	69.11	74.71
CORAL	73.42	63.88	95.32	66.70	74.83
SagNet	73.87	62.69	97.33	68.04	75.48
SelfReg	74.02	62.91	96.74	66.47	75.04
SAGM	70.97	63.56	96.81	69.33	75.17
M²-CL	**73.23**	**64.12**	**98.23**	**75.53**	**77.78**

Table 3. VLCS results (ResNet-18, top-1 accuracy %).

5.3 Office-Home

Method	Art	Clipart	Product	Real	Avg
ERM	57.45	51.74	68.11	58.27	58.89
CORAL	58.09	52.92	69.17	60.19	60.09
SagNet	57.12	52.28	69.34	60.05	59.70
SAGM	57.92	52.65	68.95	59.12	59.66
M²-CL	**58.21**	**54.02**	**71.03**	**69.82**	**63.27**

Table 4. Office-Home results (ResNet-18, top-1 accuracy %).

5.4 NICO

Method	N = 3	N = 5	N = 7
ERM	66.77	64.11	59.10
CORAL	66.45	64.39	59.43
SagNet	66.12	64.77	60.22
SAGM	66.77	65.00	59.10
M²-CL	**68.43**	**65.87**	**62.19**

Table 5. NICO results (ResNet-18, top-1 accuracy %). N = number of held-out contexts.

6. Ablation Study

We ablate each component of M²-CL on PACS and VLCS. Results confirm that parallel pipelines, spatial dropout, and the contrastive loss all contribute positively. The reduction ratio r = 4 is optimal, and loss weight $\alpha = 0.01$ is critical — large values ($\alpha = 1.0$) collapse performance by over 20%.

Configuration	PACS	VLCS
Full M²-CL (with contrastive loss)	83.54	77.78
M² only (no contrastive loss)	82.83	76.11
Cascading pipelines	80.12	73.45
Without spatial dropout	80.89	73.18
r = 2	81.73	75.34
r = 6	81.44	74.98
$\tau = 0.1$	81.67	76.02
$\tau = 2.0$	83.31	77.61
α = 10[redacted]	82.80	76.08
$\alpha = 1.0$	**63.54**	**57.22**

Table 6. Ablation study results.

7. Conclusion

M²-CL demonstrates that leveraging multi-scale and multi-layer feature representations significantly improves domain generalization. The extraction blocks with parallel concentration pipelines and spatial dropout efficiently capture domain-invariant information at multiple levels of abstraction. The multi-layer contrastive loss provides strong regularization without requiring additional data or complex training procedures. M²-CL achieves state-of-the-art results on all four standard benchmarks (PACS, VLCS, Office-Home, NICO), demonstrating consistent improvements of up to +3.61% over prior methods.

References

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